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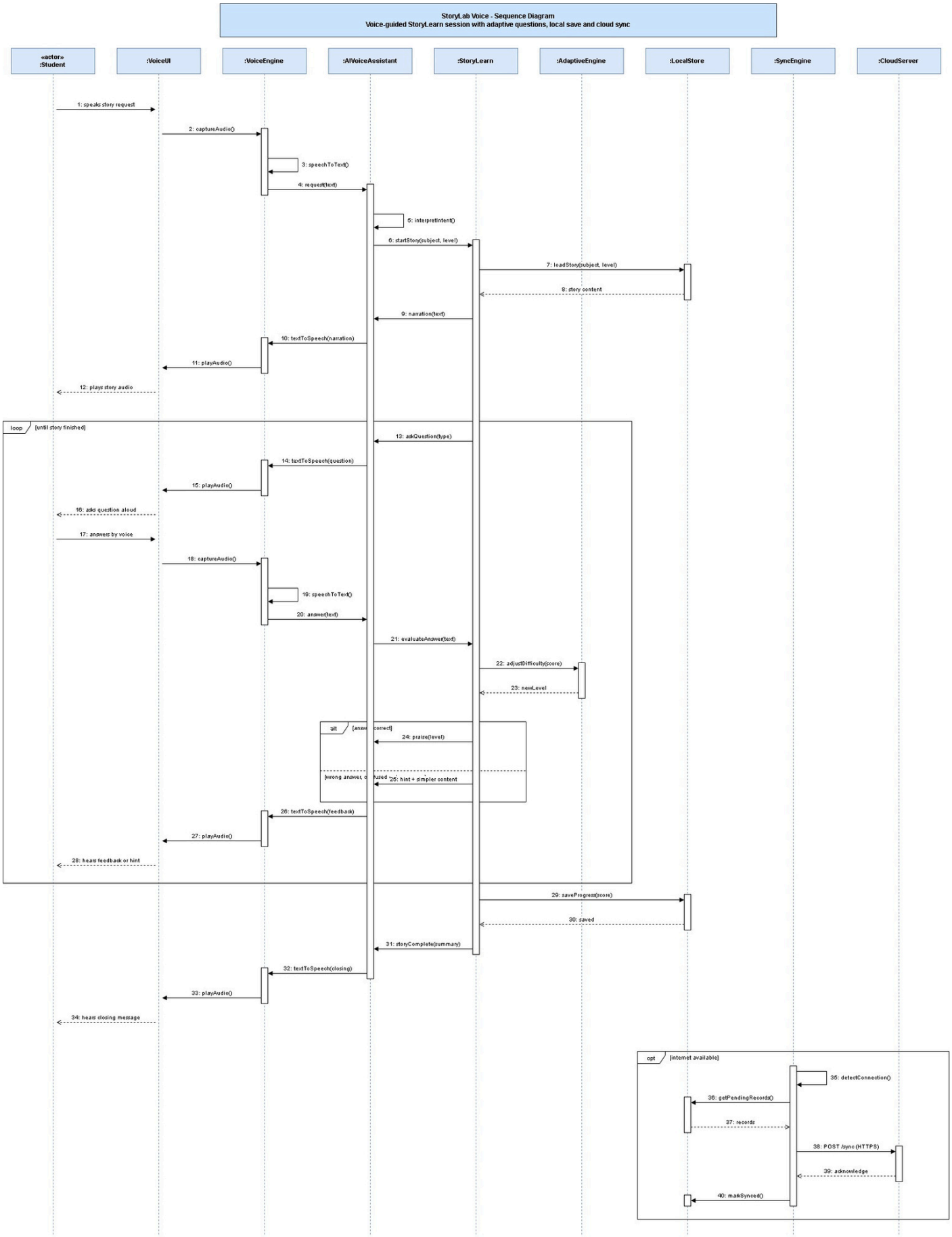
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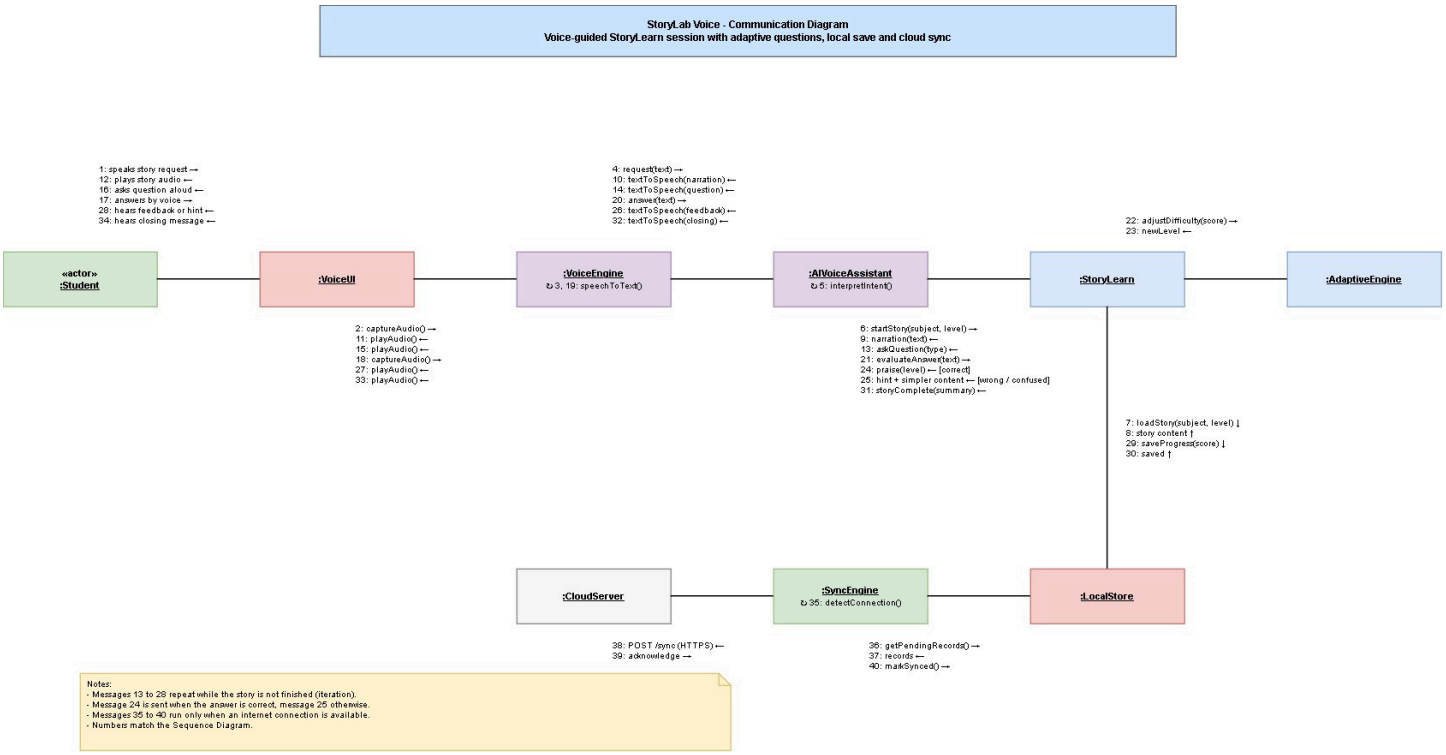
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PRACTICAL NO.06 Understanding and creating sequence diagram and communication diagram for real world application.

Sequence diagram:



Communication Diagram:



Sequence diagram: voice-guided StoryLearn session

Purpose. It shows, in time order, how the objects of StoryLab Voice exchange messages when a student learns through a story. It covers the voice pipeline, the adaptive question loop, local saving and the cloud sync.

Participants (lifelines)

Lifeline	Role
:Student	The actor, who speaks and listens
:VoiceUI	Microphone and speaker interface with the animated avatar and waveform
:VoiceEngine	Speech-to-text (STT) and text-to-speech (TTS)
:AIVoiceAssistant	The central tutor. It interprets requests and routes them.
:StoryLearn	The story engine. It narrates, asks questions and evaluates answers.
:AdaptiveEngine	Adjusts story difficulty
:LocalStore	Offline storage (IndexedDB)
:SyncEngine	Background synchronization
:CloudServer	Cloud server that receives the student's progress

Flow

Phase	Messages	What happens
Request	1–5	The student speaks a story request. VoiceUI passes the audio to VoiceEngine, which converts it to text. The assistant interprets the intent.
Story start	6–12	The assistant asks StoryLearn to start a story. StoryLearn loads it from LocalStore. The narration is converted to speech and played to the student.
Question loop	13–28	StoryLearn asks a question, which is spoken to the student. The student answers by voice, the answer is converted to text, and StoryLearn evaluates it. AdaptiveEngine adjusts the difficulty and returns the new level.
Feedback	24 or 25	If the answer is correct, StoryLearn sends praise (24). Otherwise it sends a hint with simpler content (25). The feedback is spoken to the student (26–28).
Completion	29–34	StoryLearn saves the score to LocalStore, tells the assistant the story is complete, and the closing message is spoken.
Sync	35–40	If the internet is available,

		SyncEngine reads the pending records from LocalStore, posts them to CloudServer over HTTPS, receives an acknowledgement and marks the records as synced.

Notation

- **Solid arrows** are calls, and **dashed arrows** are returns or responses.
- **Activation bars** on a lifeline show when that object is active.
- **Self-arrows** are an object calling itself: `speechToText()` (3, 19), `interpretIntent()` (5) and `detectConnection()` (35).
- **Fragments:**
  - `loop [until story finished]` covers 13–28.
  - `alt [answer correct]` and `[wrong answer, confused or long pause]` cover 24 and 25.
  - `opt [internet available]` covers 35–40.

Requirements covered: FR-CB-01 and FR-CB-02 (voice conversation), FR-CB-03 (hints when confused), FR-CB-04 (supportive feedback), FR-SL-01 and FR-SL-02 (story narration and questions) and FR-SL-03 (adaptive difficulty). The offline-first requirement shows up as the local save followed by an optional sync.

Communication diagram: the same scenario

Purpose. It shows the same interaction as the sequence diagram, but the focus is on **which objects are linked**, not on time order. The message numbers (1–40) are the same, so each message can be matched to the sequence diagram.

Objects and links. The objects are the same nine as the lifelines above. Seven links connect them along the main path:

- Student – VoiceUI
- VoiceUI – VoiceEngine
- VoiceEngine – AIVoiceAssistant
- AIVoiceAssistant – StoryLearn
- StoryLearn – AdaptiveEngine
- StoryLearn – LocalStore
- LocalStore – SyncEngine

An eighth link joins SyncEngine and CloudServer.

What each link carries

Link	Messages (→ and ← show direction)
Student – VoiceUI	1 → speaks request; 12 ← story audio; 16 ← question; 17 → answer; 28 ← feedback; 34 ← closing message
VoiceUI – VoiceEngine	2 and 18 → captureAudio(); 11, 15, 27 and 33 ← playAudio()
VoiceEngine – AIVoiceAssistant	4 and 20 → request or answer as text; 10, 14, 26 and 32 ← textToSpeech()
AIVoiceAssistant – StoryLearn	6 → startStory; 21 → evaluateAnswer; 9, 13, 24, 25 and 31 ← narration, question, praise, hint and completion
StoryLearn – AdaptiveEngine	22 → adjustDifficulty; 23 ← newLevel
StoryLearn – LocalStore	7 and 29 ↓ load and save; 8 and 30 ↑ replies
SyncEngine – LocalStore	36 and 40 → read pending records and mark synced; 37 ← records
SyncEngine – CloudServer	38 → POST /sync over HTTPS; 39 ← acknowledgement

Self-calls (`speechToText()`, `interpretIntent()`, `detectConnection()`) are written inside the object boxes with a ↻ symbol. A note at the bottom records the loop (13–28), the alternative (24 or 25) and the optional sync (35–40).

Sequence vs communication. The sequence diagram emphasizes *when* each message happens. The communication diagram emphasizes *who talks to whom*. It makes the architecture visible: the assistant sits in the middle, the voice pipeline runs on its left, and the story, adaptive and storage objects run on its right.